

Invoice Canonicalization Agent

Deterministic first. AI only for uncertainty.
Human approvals become cheaper future knowledge.

TIER 1

Approved lookup

no LLM

TIER 2

Retrieval / AI

bounded

TIER 3

Human learning

governed

3 / 3

runs converge

18 / 18

observed variants

0

LLM calls

The brief shows three different runs. All three must converge.

We turned every supplied description into an executable acceptance contract: 3 runs × 6 products = 18 observed variants.

#	RUN 1 observed	RUN 2 observed	RUN 3 observed	REQUIRED RESULT
1	Steel Accessories	Highlife Components	Steel Fittings	Highlife Components
2	Unstoppable Sneakers	Performance Sneakers	Athletic Shoes	Performance Sneakers
3	Polarbear T-Shirt	White Tee	Logo T-Shirt	White Tee
4	Grizzly T-Shirt	Beige Tee	Logo Tee	Beige Tee
5	El Camino Shorts	Casual Shorts	Quick-Dry Shorts	Casual Shorts
6	Black Socks	Crew Socks	Athletic Crew Socks	Crew Socks

Acceptance rule: every supplied variant resolves to the same tenant-scoped canonical product ID - by approved exact lookup, not generation.

The actual invoice input also resolves 6/6 deterministically

The raw descriptions on the supplied invoice map directly to the requested canonical taxonomy with approved aliases and zero model calls.

INVOICE

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01. Juli 2025

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Due in 15 days

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DESCRIPTION	QTY	UNIT PRICE	TOTAL
Highlife Steel Accessories	1	10	10.00
Sneaker "Unstoppable"	2	11	22.00
T-Shirt White "Polarbear"	3	12	36.00
T-Shirt Beige "Grizzly"	4	13	52.00
Shorts "El Camino"	5	14	70.00
Socks, black	6	15	90.00
SUBTOTAL			280.00
DISCOUNT			13.00
SUBTOTAL LESS DISCOUNT			267.00
TAX RATE			20.00%
TOTAL TAX			53.40
SHIPPING/HANDLING			12.00
Balance Due			\$ 332.40

Remarks / Payment Instructions:

Company Signature

Client Signature

Deterministic replay

Highlife Steel Accessories	→ Highlife Components	NO LLM
Sneaker "Unstoppable"	→ Performance Sneakers	NO LLM
T-Shirt White "Polarbear"	→ White Tee	NO LLM
T-Shirt Beige "Grizzly"	→ Beige Tee	NO LLM
Shorts "El Camino"	→ Casual Shorts	NO LLM
Socks, black	→ Crew Socks	NO LLM

6 / 6**raw invoice mappings**

deterministic replay

0**LLM calls**

challenge path

PASS**extraction gate**

6/6 row equations

280=280**subtotal check**

Decimal

Together with the 18/18 three-run acceptance matrix, this proves the supplied task end-to-end. Unseen-model quality is evaluated separately.

What the task actually requires

A reproducibility problem first: the same booking text should resolve to the same canonical result across repeated runs and client contexts.

ISSUES IN THE BRIEF

- same invoice text should map to one reproducible description
- direct LLM generation can vary between repeated runs
- different clients and business partners can use different taxonomies
- the solution must specify how consistency is enforced

DESIGN RESPONSE

- approved canonical IDs + tenant-scoped aliases
- deterministic exact-match path before any AI call
- bounded AI only for unseen or uncertain descriptions
- human approval promotes trusted future exact aliases
- guardrails + regression tests make behavior auditable

DESIGN PRINCIPLE Determinism is a data and governance property - not a prompt-engineering trick.

Step 1 - validate invoice evidence before canonicalization

Known or wild, the same extraction and arithmetic quality gate runs first. No product enters canonicalization from dirty invoice evidence.



Every line — tamed or wild — passes the same extraction and arithmetic gate.

quantity × unit price = line total · Σ line totals = subtotal · discount / tax / shipping reconciled separately

Only clean product-line evidence moves forward into deterministic lookup or the uncertainty path.

Step 2 - known products take the fast lane; unknowns are routed deliberately

Approved invoices are deterministic; unseen descriptions are investigated, not silently guessed.



Known / Tamed

Exact approved alias → stable canonical product ID. No LLM call.



Unknown / Wild

Fast lane blocked. Unknown is routed deliberately — not treated as a failed invoice.



Approved knowledge first

Before AI, the system checks approved canonical knowledge and past human decisions.

Step 3 - retrieval, bounded AI, and human approval resolve the unknown

Tier 2 gathers evidence and proposes a candidate. Tier 3 approves knowledge before anything becomes trusted.



Retrieve evidence → bounded AI proposal → human approves / edits / rejects

An unapproved AI guess never becomes trusted knowledge. Human approval is the promotion step.

Step 4 - approval turns uncertainty into reusable deterministic knowledge

Once approved, the same description becomes an exact alias match on future invoices.



0

LLM calls on future exact matches

0

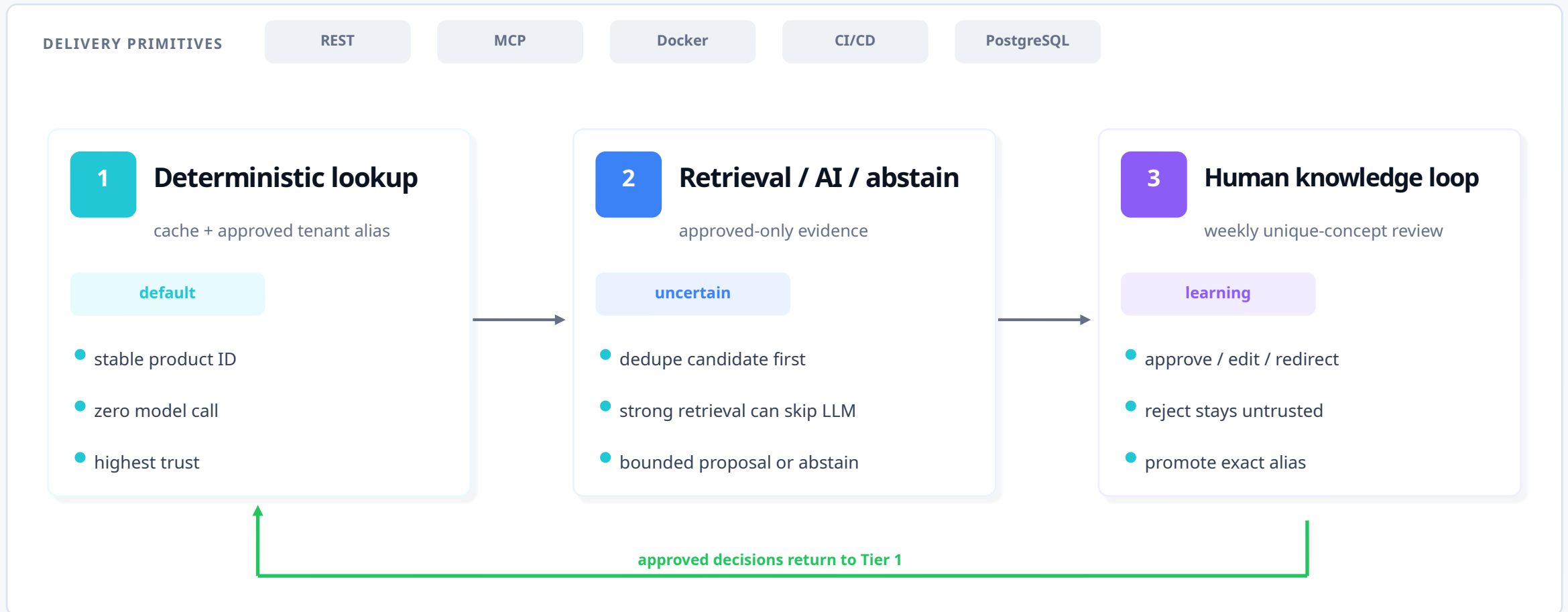
human reviews on future exact matches



marginal model cost as the alias registry grows

Three-tier canonicalization architecture

That story maps directly to three production tiers; delivery primitives stay outside the business logic.



Story -> code: Approved Lookup = approved aliases | Uncertainty Lab = approved retrieval + bounded AI / abstention | Human Approval = review queue -> knowledge promotion.

Measured reliability: verified gates, not vanity metrics

The final code now passes static checks and keeps golden-set evidence separate from production-accuracy claims.

111**automated tests**

offline suite

18/18**challenge variants**

3 runs converge

11/11**routing cases**

curated regression

PASS**Ruff + mypy**

static checks

0**unsafe auto-accepts**

target 0

What the results prove

- all 3 supplied runs converge to the same six canonical descriptions
- 26 repeated unknown observations collapse to 1 proposal
- 0 unexpected LLM calls on the golden regression set
- 0 model-review bypasses and unsafe inputs can abstain
- 83% branch-aware coverage and package installs in isolation

What we do NOT claim

- 18/18 challenge acceptance = ML accuracy (it is deterministic alias coverage)
- 11 curated cases predict production accuracy at scale
- vector RAG outperforms lexical retrieval (not benchmarked)
- offline fixtures validate real provider latency / cost / drift

Current verified gate: 111 tests · 83% branch-aware coverage · 18/18 supplied variants · Ruff PASS · mypy PASS · 11/11 curated routing cases · 0 unsafe auto-accepts.

TAKEAWAY

Reproducible by default. AI only where uncertainty remains.

The system solves the challenge deterministically, then uses bounded AI only for the unknown tail.

DETERMINISTIC

Approved aliases return the same canonical product every time.

GOVERNED

Unapproved AI proposals never become trusted knowledge silently.

LEARNING

Human approval converts uncertainty into future exact matches.

“The goal is not to make the LLM more consistent. It is to need the LLM less often.”

18/18 supplied variants · 6/6 raw invoice · 11/11 golden routing · Ruff PASS · mypy PASS · 111 tests

APPENDIX

Technical depth when the interviewer asks

Unknown lifecycle, review workflow, tenant isolation, delivery packaging, testing, retrieval, security, CI/CD, and limitations.

18-stage quality contract

test matrix

security + cost

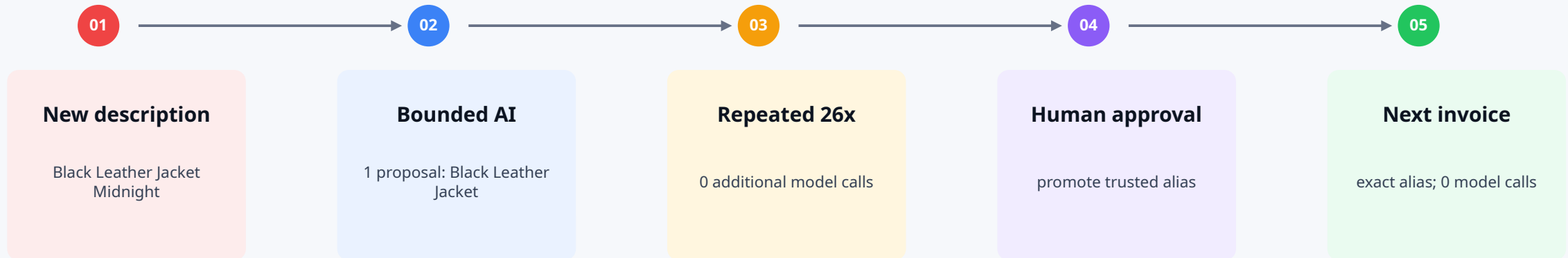
retrieval / RAG

review data model

limitations

An unseen description uses AI once - then becomes deterministic

The same pending concept is reused before approval; approval converts it into exact knowledge.



Scalability rule

The model is called once per unique pending concept - not once per invoice row.

This is the declining cost curve in operational form.

**26
observations
1 proposal**

**after approval
0 future calls**

Humans approve knowledge - not invoices

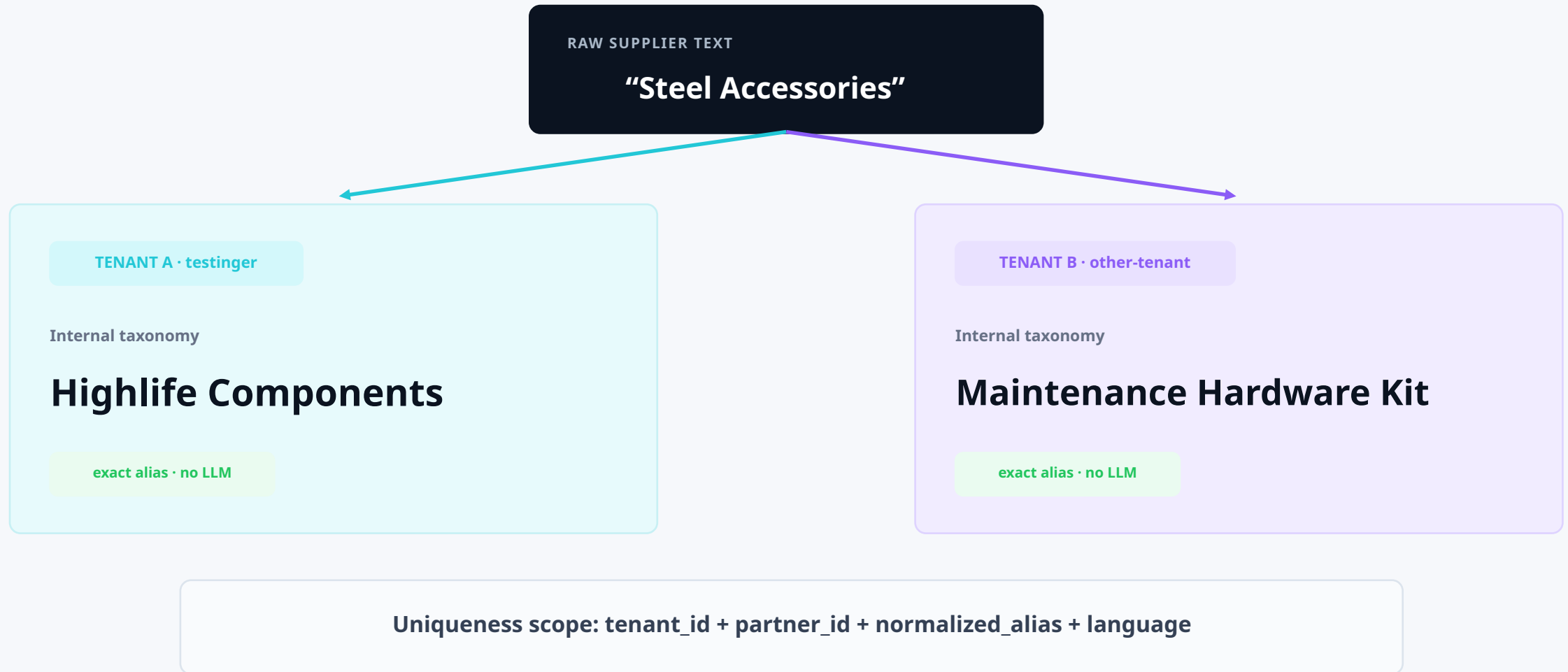
A weekly queue contains unique unresolved concepts with evidence, impact, and flexible actions.

PENDING KNOWLEDGE REVIEW						
				27 unique concepts	71,482 lines affected	
Priority	Observed variants	Proposal	Score	LLM	Occurrences	Status
HIGH	Air Runner Pro Black AirRunner Pro blk	Performance Sneakers	98.2%	Yes	18,400	approve
HIGH	Black Leather Jacket Midnight	Black Leather Jacket	73.0%	Yes	26	waiting
MED	Athletic crew socks	Crew Socks	97.8%	No	127	waiting
Reviewer actions						
Approve existing Redirect Create new Edit + approve Reject Defer						

Completed decisions leave the active queue but remain archived + audited.

Tenant taxonomies prevent cross-client collisions

The raw string is not a global key: alias lookup is scoped by tenant + partner + language.



Authentication binds the request to a tenant; PostgreSQL RLS is a second defense-in-depth layer.

Core solution vs. production packaging

This avoids over-engineering the story: the business logic stays small; the rest is delivery discipline.

CORE BUSINESS LOGIC

- canonical product IDs
- tenant-scoped approved aliases
- normalization + retrieval policy
- bounded AI for unknowns
- human knowledge promotion

OPERATIONAL GUARDRAILS

- extraction arithmetic gate
- auth + tenant binding
- budgets / timeouts / retries
- validation + abstention
- audit + monitoring

DELIVERY PRIMITIVES

- REST API
- MCP integration boundary
- Docker / Compose
- GitHub CI/CD
- PostgreSQL migrations

Interview rule: lead with the first column. Use the other two only when asked how you would productionize it.

Internal quality sequence - 18 machine-checked stages

Appendix detail: this is an implementation contract, not the primary business architecture.

01-06

Input + evidence

- auth + tenant binding
- file validation
- deterministic parsing
- Decimal line gate
- model extraction fallback
- persist invoice envelope

07-10

Deterministic state

- normalize text
- version-safe cache
- approved exact alias
- atomic pending claim

11-14

Uncertainty handling

- approved-only retrieval
- policy scoring
- bounded generation
- output/security validation

15-18

Governance

- staged review queue
- human action
- knowledge promotion
- audit + metrics

Testing strategy

Offline determinism for CI; live-provider evaluation remains opt-in.

Layer	What it proves	Property
Unit / contracts	normalization, money, budgets, prompt schemas	fast / deterministic
Integration	repository, canonicalization, review promotion	real application wiring
Document metamorphic	PDF / XLSX / DOCX / JSON / CSV / TXT	same business result across formats
AI fixtures	stored prompt examples + model outputs	no API-spend CI, stable regressions
Adversarial	prompt injection, PII/financial leakage, cross-tenant, malformed output	safe failure / abstention
Concurrency	same unknown arriving simultaneously	one candidate / one proposal
Packaging	wheel build + isolated install smoke	delivery integrity

Challenge acceptance: 3 supplied runs × 6 descriptions = 18/18 variants → the same 6 canonical outputs, with 0 LLM calls.

Security, reliability, and cost guardrails

These protect the product logic; they do not decide the canonical product.

SECURITY

- tenant-bound auth
- PostgreSQL RLS
- party PII kept out of prompts
- prompt-injection guard
- CSV formula neutralization

RELIABILITY

- Decimal financial fields
- line arithmetic gate
- commercial reconciliation
- schema + attribute validation
- immutable audit history

COST / LATENCY

- exact alias before retrieval
- dedupe before model
- max calls + dollar budget
- timeouts + bounded retries
- cache by taxonomy version

Retrieval / RAG boundary

Use the simplest retrieval method that meets false-match, latency, and recall targets.

CURRENT ASSESSMENT

- approved aliases only
- sequence similarity
- token overlap
- character trigrams
- category + protected attributes
- margin vs. second candidate

SCALE-UP OPTION

- SemanticScoreProvider interface exists
- PostgreSQL schema is pgvector-ready
- benchmark Recall@K + false match first
- measure latency + memory before enabling
- do not index unapproved model suggestions

Anti-feedback-loop rule: staging data never becomes retrieval evidence until a human promotes it.

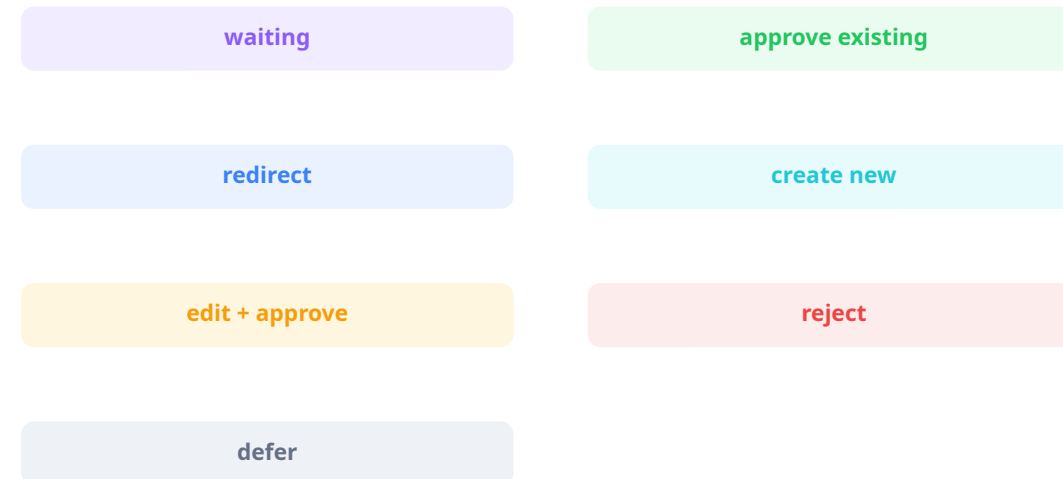
Review data model and state transitions

Local CSV is an editable projection; production persistence is relational.

Candidate record

- normalized candidate key
- source variants + occurrence count
- affected value by currency
- retrieval / policy scores
- model + prompt provenance
- risk flags + evidence
- human overrides + notes

Human-editable final status



Processed rows leave the active work queue but remain in archive + audit history.

Concurrency: one active candidate per tenant + partner + normalized description.

Limitations and next validation

What still needs evidence before a real production rollout.

Area	Current evidence	Next validation
Production accuracy	11 curated cases are regression evidence, not population statistics	Build a blind labelled dataset across clients/categories/languages
Threshold calibration	current thresholds are configuration, not statistically calibrated probabilities	Optimize to a business target such as false auto-resolution < 0.1%
Live provider	offline fixtures prove contracts but not real latency/cost/drift	Benchmark candidate models on a blind labelled set; pin the best quality / cost / latency trade-off
Semantic retrieval	pgvector seam exists but is not benchmarked as better	Compare Recall@K / latency / false-match against lexical hybrid
Release environment	Ruff + mypy pass; Docker remains a separate release-environment check	Keep Docker build / smoke test as an independent release CI job
Operational UX	CSV proves workflow but is not a polished reviewer product	Replace with dashboard when reviewer volume justifies it

Delivery and demo commands

Everything needed to inspect, test, demonstrate, and package the solution.

<code>./run_assessment.sh</code>	Offline quality gate: tests, coverage, evaluation, package smoke
<code>make interview-demo</code>	Challenge 3-run acceptance + three-tier lifecycle + machine-readable evidence
<code>make review-demo</code>	Generate the editable pending-knowledge CSV
<code>make review-process</code>	Apply reviewer decisions, archive completed items
<code>make assess-full</code>	Strict release gate including Ruff, mypy, Docker

Assessor-facing artifacts

- README.md
- docs/assessor_summary.md
- docs/invoice_data_model.md
- reports/final_delivery_check.md
- docs/verified_results.md
- docs/production_validation_plan.md
- presentation/PPTX + PDF

Version 1.2.0